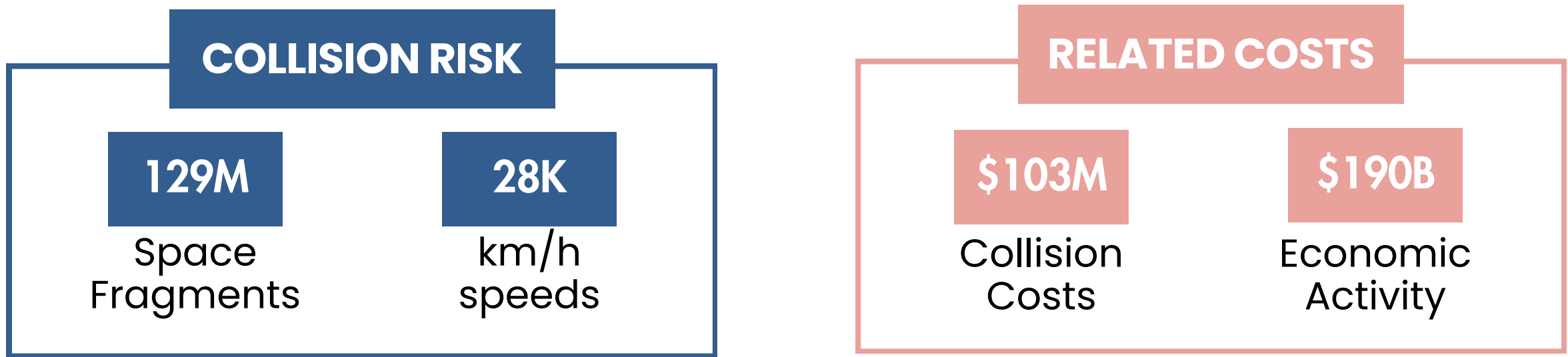


# VISUALIZING SPACE DEBRIS

## TEAM 9: LITTLE EINSTEINS

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### MOTIVATION & INTRODUCTION

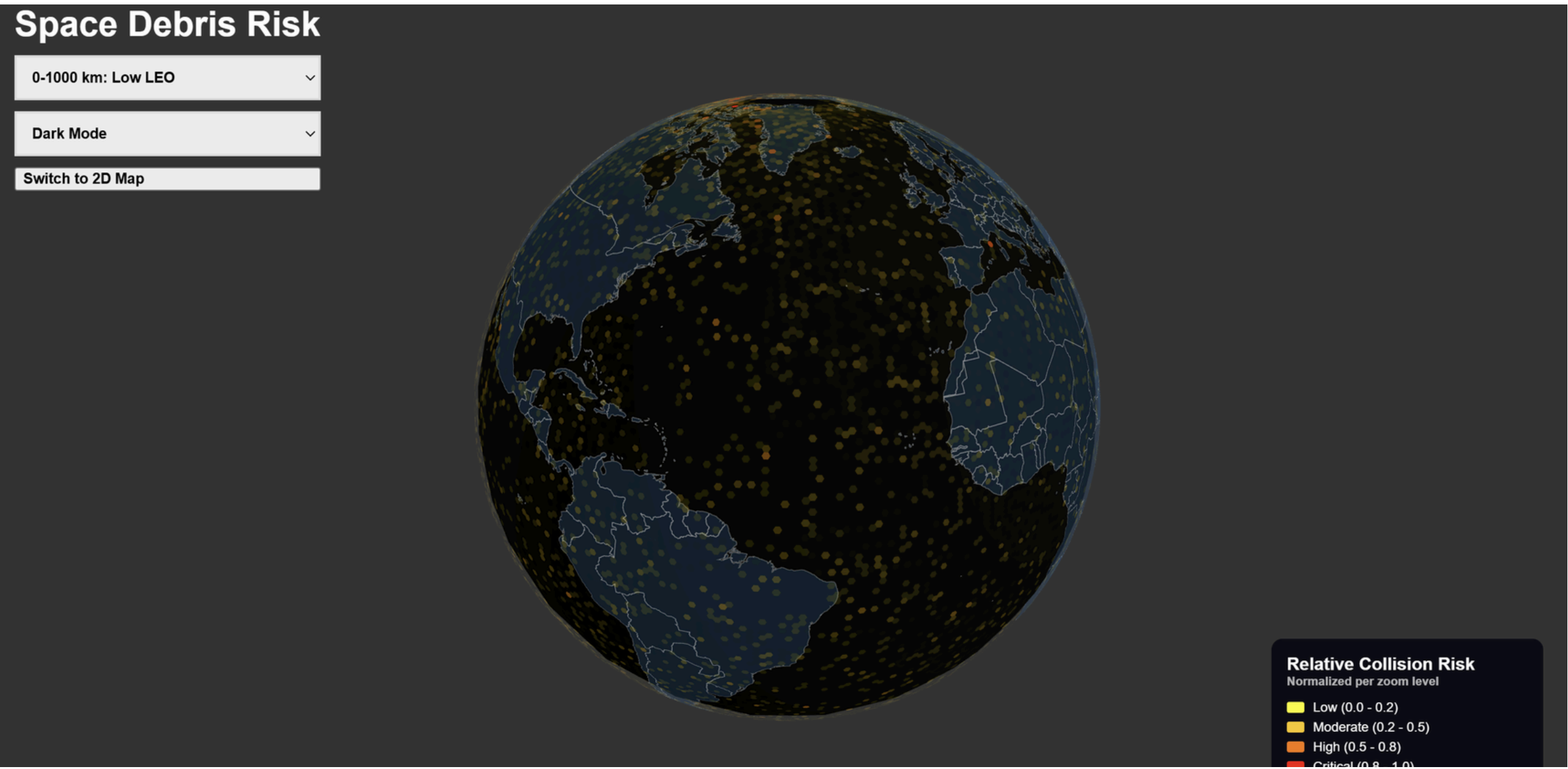
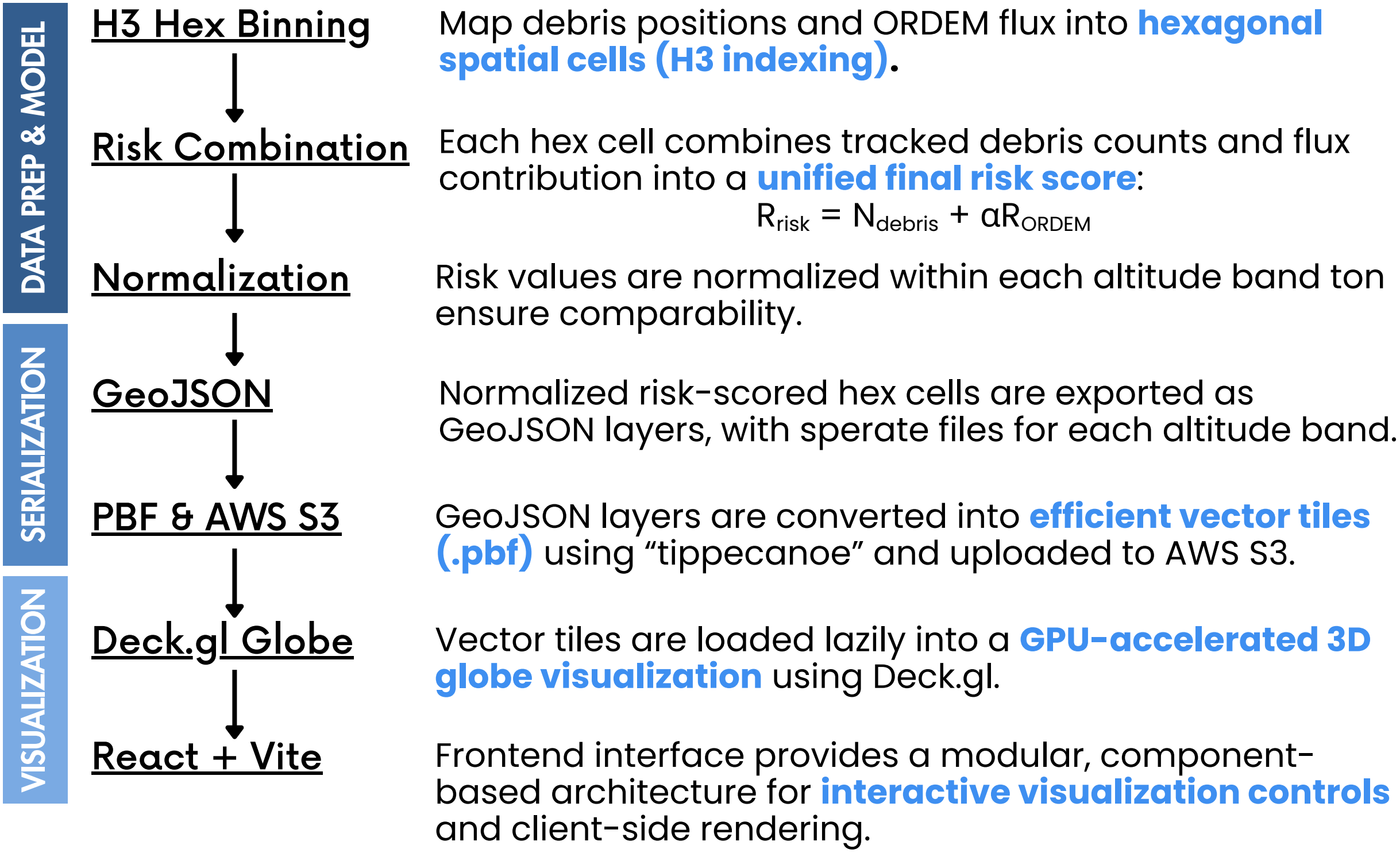


This project addresses gaps in existing approaches by fusing statistical flux data with tracked debris, delivering an interactive 3D probabilistic heatmap of the **full debris risk landscape** accessible to technical and non-technical stakeholders alike.

### OUR APPROACH

#### Approach & Technical Implementation

We developed a unified spatial risk model and visualization pipeline from our data sources. This system transforms raw data into an interactive risk visualization.



### RESULTS

#### How We Evaluated

- 80/20 train-test split on satellite data to avoid bias
- Compared predicted risk zones against 3,233 held-out test points
- Measured rendering performance across 4 runs using Chrome DevTools

#### Model Accuracy

- Overall hit rate: **97.5%** (3,151/3,233 test points)
- LEO performance: **~97%** across both LEO altitude bands
- MEO performance: 94% in altitudes 2000–30,000 km
- GEO performance: 98% in altitudes 30,000 km+
- ~54%** of debris falls in top 20% highest-risk cells (2x better than random)
- Risk scores are heavily **right-skewed** with median of 0.03 vs. mean of 0.08. This means that most regions are low-risk with a small number of high-danger hotspots driving the average up.

#### Rendering Performance

| Interaction      | FPS Range |
|------------------|-----------|
| Zoom In          | 3–35 FPS  |
| Zoom Out         | 3–24 FPS  |
| Rotation Up/Down | 5–40 FPS  |
| Rotation Left    | 6–38 FPS  |
| Rotation Right   | 3–40 FPS  |

- FPS ranges 3–40 across all interaction types
  - Rotation performs smoothest (up to 40 FPS)
  - Zoom operations show more variability (3–24 FPS)
- GPU memory usage: 2.9–8.6 MB — **low overhead**
- Avg initial load time:
  - Chrome 6.16s | Brave 1.27s | Safari 0.49s
- Load time variance driven by browser caching, not rendering complexity.

Overall, the visualization demonstrates **efficient browser-based rendering** with **minimal GPU overhead**, confirming that our heatmap approach is a scalable alternative to traditional point-based debris visualization.

### DATA

| NASA ORDEM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | CELESTRAK + SPACETRACK                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>About</b> <ul style="list-style-type: none"><li>NASA statistical model which estimates orbital debris flux across altitude bands and particle size.</li></ul> <b>Retrieval</b> <ul style="list-style-type: none"><li>Downloaded from NASA’s web application via batch computations.</li></ul> <b>Processing</b> <ul style="list-style-type: none"><li>Combined flux tables, aggregated particle-size flux into single risk value, and interpolated data.</li><li>~52 million flux points (1750 MB).</li></ul> | <b>About</b> <ul style="list-style-type: none"><li>Provides TLE orbital parameters for tracked objects in space.</li></ul> <b>Retrieval</b> <ul style="list-style-type: none"><li>Obtained through API endpoints, which allows us to access TLE sets.</li></ul> <b>Processing</b> <ul style="list-style-type: none"><li>Propagated TLEs with SGP4 to estimate timestamped latitude, longitude, and altitude values.</li><li>~16,000 tracked debris objects (3.51 MB disk space).</li></ul> |

#### Innovation & Impact

Our approach overcomes **memory, CPU, and browser rendering bottlenecks** by replacing traditional point-level rendering with an aggregate, tile-based heatmap pipeline.

#### Performance Improvements

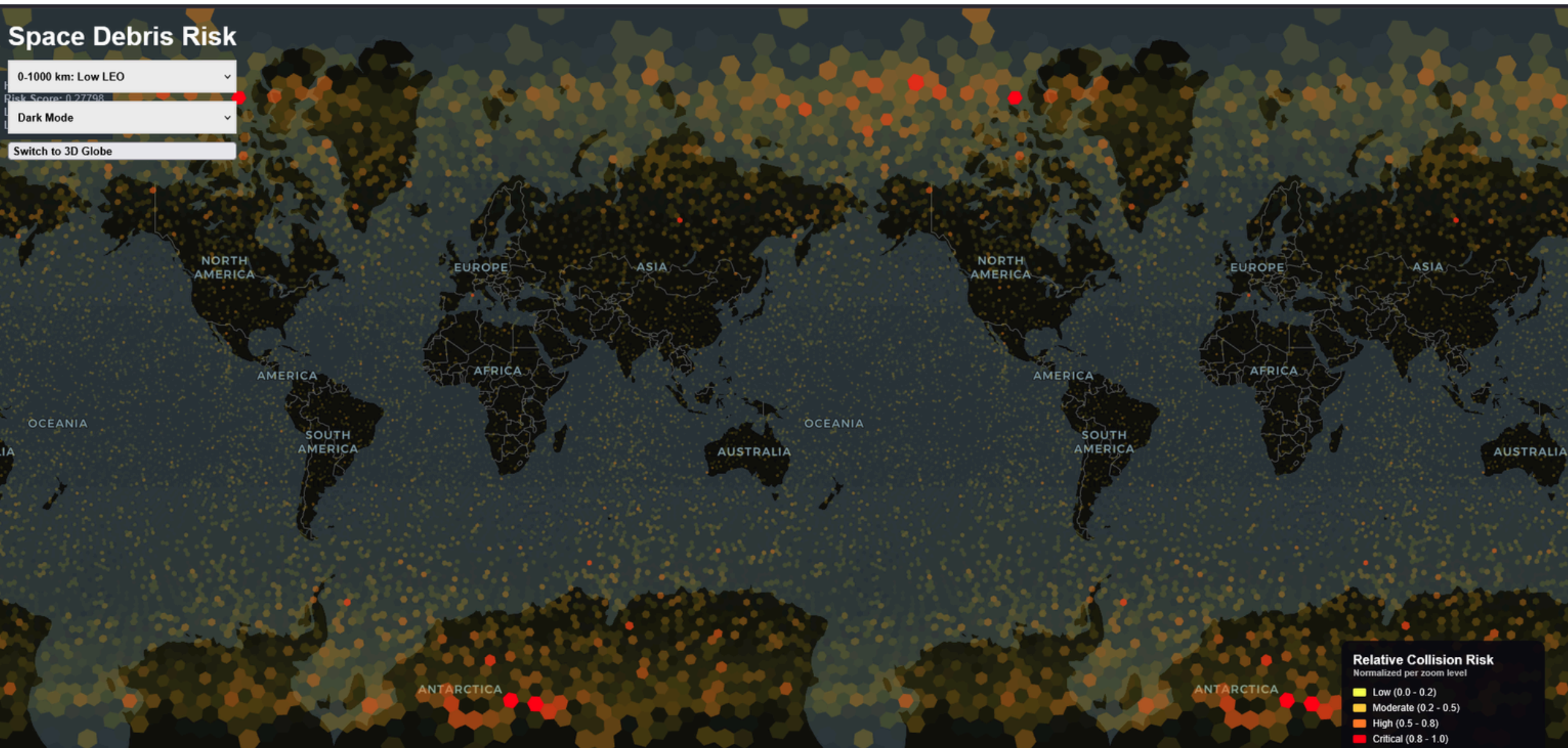
- H3 hex binning** aggregates raw data points from our two sources into more scalable spatial cells.
- Dynamic resolution scaling** increases or decreases hex detail as the user zooms in and out.
- Lazy loading** renders only the visible onscreen tiles, saving significant compute.
- Precomputed vector tiles** remove expensive real-time computation.
- AWS S3 bucket storage** enables quick and easy access to precomputed vector tiles, allowing Deck.gl to request only the specific tile endpoints needed for efficient rendering.

#### Holistic Environment View

By combining tracked debris and modeled flux estimates, we produce a more complete representation of the orbital debris environment, including both:

- Observed large debris
- Statistically modeled smaller debris

Our approach shifts debris visualization from object tracking to spatial risk estimation while simultaneously improving the scalability and interpretability of the visualization.



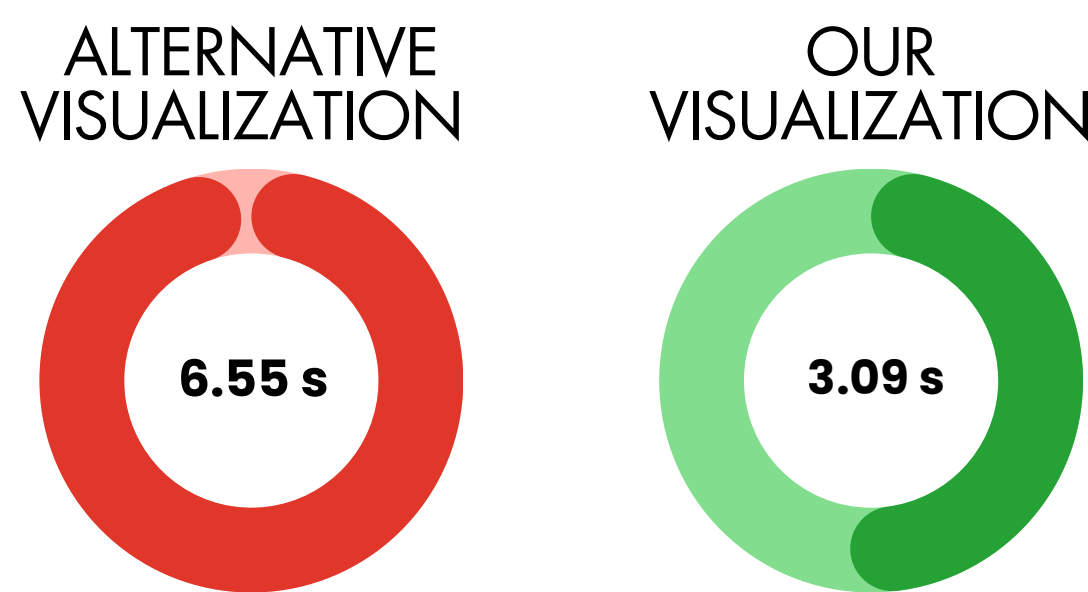
### COMPARING EXPERIMENTS

Our method is unique in that it uses **both probability density estimation and both an interactive 2D and 3D heat map visualization**, as opposed to using object tracking techniques or producing static statistical outputs.

| Other Approaches                   | Key Limitations            | Our Visualization               |
|------------------------------------|----------------------------|---------------------------------|
| Point-based deterministic plotting | Breaks down at scale       | Covers 50M+ untracked fragments |
| ORDERM 3.0 / MASTER-2009           | Static, non-spatial output | Explorable by altitude/region   |
| Alfano collision risk              | Object-level, not scalable | Population-level coverage       |

- Point-based approach** is not possible beyond a couple thousand objects. However, our model takes into account **50 million** points of data, between **1mm–10cm** in size, which are too small to be detected using current tracking methods.
- ORDERM 3.0 / MASTER-2009** produce static, non-spatial outputs, in contrast to our fully interactive and explorable visualization by altitude band and region.
- Alfano collision risk models** operate at the object level and don’t scale, while our approach provides population-level coverage across all orbital bands simultaneously.

#### We also conducted a user test for usability and interpretability:



Our visualization was **52%** faster in analyzing debris around the South Pole. Users agreed that our visualization was clean and intuitive, unlike the alternative tool, which was described as laggy and cluttered.